

## VOLGOGRAD GUMRAK, Russian Federation

WMO: 345600

Lat: 48.783N

Lon: 44.346E

Elev: 147

StdP: 99.57

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -22.4      | -19.5 | -25.4                         | 0.4 | -22.2 | -22.2 | 0.5 | -19.1 | 13.4                  | -4.0 | 12.3 | -4.8 | 4.2                   | 70   | 0.641 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.7                   | 35.8            | 18.8 | 33.8 | 18.7 | 31.9 | 18.3 | 20.8                | 29.8 | 20.1 | 29.2 | 19.4 | 28.4 | 5.2                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.2                            | 13.3 | 23.0 | 17.2 | 12.5 | 22.7 | 16.2 | 11.7 | 22.4 | 60.6          | 29.9 | 58.1 | 29.0 | 55.8 | 28.2 | 24.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 12.0          | 10.6 | 9.6 | -24.2                      | 37.8 | 4.5                | 1.9 | -27.4                                              | 39.1 | -30.0      | 40.2 | -32.6      | 41.3 | -35.8      | 42.7 |    |    |
| (5)               |      |     | -24.4                      | 22.7 | 4.4                | 1.0 | -27.5                                              | 23.4 | -30.1      | 24.0 | -32.6      | 24.5 | -35.7      | 25.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.8    | -6.7  | -6.0  | 0.3   | 9.8   | 16.6  | 21.4  | 24.1  | 23.2  | 16.4  | 8.7   | 1.1   | -4.2  |  |  |
|                                                                        | DBStd       | 12.20  | 6.52  | 6.53  | 5.06  | 4.54  | 4.55  | 4.23  | 3.56  | 4.14  | 4.55  | 4.97  | 5.30  | 5.90  |  |  |
|                                                                        | HDD10.0     | 2128   | 517   | 448   | 301   | 59    | 5     | 0     | 0     | 0     | 5     | 85    | 269   | 439   |  |  |
|                                                                        | HDD18.3     | 3994   | 776   | 682   | 558   | 258   | 88    | 17    | 2     | 8     | 89    | 301   | 518   | 697   |  |  |
|                                                                        | CDD10.0     | 1695   | 0     | 0     | 1     | 52    | 210   | 343   | 438   | 410   | 196   | 44    | 2     | 0     |  |  |
|                                                                        | CDD18.3     | 518    | 0     | 0     | 0     | 1     | 36    | 111   | 181   | 159   | 30    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 5747   | 0     | 0     | 0     | 13    | 377   | 1224  | 2028  | 1822  | 274   | 9     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 2506   | 0     | 0     | 0     | 1     | 112   | 522   | 943   | 854   | 72    | 2     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.9    | 5.3   | 5.7   | 5.6   | 5.1   | 4.6   | 4.3   | 4.1   | 4.2   | 4.4   | 4.6   | 4.9   | 5.3   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 378    | 34    | 28    | 29    | 26    | 43    | 43    | 27    | 23    | 31    | 30    | 29    | 38    |  |  |
|                                                                        | PrecMax     | 529    | 67    | 61    | 57    | 87    | 95    | 120   | 76    | 66    | 104   | 98    | 80    | 95    |  |  |
|                                                                        | PrecMin     | 251    | 14    | 15    | 4     | 2     | 10    | 3     | 3     | 2     | 0     | 6     | 7     | 9     |  |  |
|                                                                        | PrecStd     | 79     | 15    | 11    | 16    | 21    | 26    | 32    | 19    | 17    | 29    | 22    | 18    | 19    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 6.6   | 7.0   | 15.2  | 25.1  | 32.2  | 36.2  | 38.0  | 37.9  | 31.9  | 24.0  | 14.6  | 8.0   |  |  |
|                                                                        |             | MCWB   | 5.2   | 5.1   | 9.0   | 14.7  | 17.1  | 18.8  | 19.0  | 19.2  | 17.7  | 15.0  | 10.4  | 7.2   |  |  |
|                                                                        | 2%          | DB     | 3.9   | 4.2   | 12.0  | 22.2  | 29.8  | 33.9  | 35.5  | 35.8  | 28.7  | 20.8  | 11.8  | 6.1   |  |  |
|                                                                        |             | MCWB   | 3.5   | 3.1   | 7.1   | 13.4  | 16.8  | 18.7  | 19.1  | 18.7  | 16.2  | 13.4  | 9.2   | 5.5   |  |  |
|                                                                        | 5%          | DB     | 2.1   | 2.8   | 9.5   | 19.9  | 27.3  | 31.9  | 33.8  | 33.4  | 26.5  | 18.1  | 9.9   | 4.6   |  |  |
|                                                                        |             | MCWB   | 1.6   | 2.0   | 5.9   | 12.1  | 16.2  | 18.4  | 18.9  | 18.4  | 15.2  | 12.4  | 8.4   | 4.1   |  |  |
|                                                                        | 10%         | DB     | 1.0   | 1.2   | 7.1   | 17.6  | 25.1  | 29.8  | 31.8  | 31.2  | 24.2  | 16.0  | 8.0   | 2.8   |  |  |
|                                                                        |             | MCWB   | 0.7   | 0.7   | 4.8   | 10.8  | 15.6  | 17.7  | 18.5  | 17.9  | 14.7  | 11.7  | 6.9   | 2.3   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 5.2   | 5.2   | 10.0  | 15.9  | 19.4  | 21.4  | 21.7  | 21.6  | 19.3  | 16.5  | 11.8  | 7.2   |  |  |
|                                                                        |             | MCDB   | 6.0   | 6.5   | 14.1  | 22.7  | 27.0  | 30.5  | 30.4  | 31.7  | 27.9  | 22.4  | 14.5  | 7.5   |  |  |
|                                                                        | 2%          | WB     | 3.3   | 3.2   | 7.9   | 14.2  | 18.4  | 20.5  | 20.8  | 20.4  | 17.9  | 14.7  | 9.9   | 5.6   |  |  |
|                                                                        |             | MCDB   | 3.7   | 4.0   | 10.9  | 20.5  | 26.3  | 29.6  | 29.5  | 29.9  | 24.4  | 18.5  | 11.1  | 6.0   |  |  |
|                                                                        | 5%          | WB     | 1.5   | 1.9   | 6.2   | 13.0  | 17.3  | 19.6  | 20.1  | 19.6  | 16.8  | 13.5  | 8.4   | 4.0   |  |  |
|                                                                        |             | MCDB   | 1.8   | 2.4   | 8.8   | 18.1  | 25.1  | 28.4  | 29.2  | 29.0  | 23.3  | 16.9  | 9.3   | 4.4   |  |  |
|                                                                        | 10%         | WB     | 0.6   | 0.8   | 4.7   | 11.6  | 16.2  | 18.7  | 19.5  | 18.8  | 15.6  | 12.1  | 6.8   | 2.1   |  |  |
|                                                                        |             | MCDB   | 0.8   | 1.2   | 6.9   | 16.5  | 23.2  | 26.9  | 28.6  | 28.4  | 22.2  | 15.4  | 7.9   | 2.5   |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 5.5   | 6.3   | 7.5   | 10.6  | 11.5  | 11.7  | 11.7  | 12.3  | 11.0  | 8.8   | 6.0   | 5.3   |  |  |
|                                                                        | 5% DB       | MCDBR  | 4.8   | 5.4   | 11.4  | 13.3  | 14.0  | 13.6  | 13.6  | 14.1  | 13.4  | 11.9  | 8.0   | 5.3   |  |  |
|                                                                        |             | MCWBR  | 4.4   | 4.6   | 7.3   | 6.6   | 5.9   | 4.8   | 4.6   | 4.7   | 5.5   | 6.4   | 6.0   | 4.9   |  |  |
|                                                                        | 5% WB       | MCDBR  | 4.6   | 5.1   | 10.1  | 11.8  | 12.3  | 12.1  | 11.8  | 12.3  | 11.5  | 10.3  | 7.0   | 5.4   |  |  |
|                                                                        |             | MCWBR  | 4.4   | 4.6   | 6.9   | 6.5   | 5.9   | 4.9   | 4.5   | 4.9   | 5.9   | 6.3   | 6.0   | 5.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.328 | 0.360 | 0.388 | 0.446 | 0.414 | 0.405 | 0.421 | 0.430 | 0.395 | 0.365 | 0.341 | 0.329 |  |  |
|                                                                        | taud        |        | 2.131 | 2.040 | 2.134 | 2.066 | 2.248 | 2.317 | 2.273 | 2.227 | 2.293 | 2.327 | 2.283 | 2.178 |  |  |
|                                                                        | Ebn at Noon |        | 691   | 754   | 810   | 800   | 847   | 858   | 835   | 808   | 799   | 758   | 684   | 635   |  |  |
|                                                                        | Edh at Noon |        | 85    | 118   | 127   | 151   | 132   | 124   | 127   | 128   | 108   | 89    | 74    | 72    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.91  | 1.84  | 3.10  | 4.56  | 6.00  | 6.54  | 6.36  | 5.63  | 4.04  | 2.32  | 1.07  | 0.64  |  |  |
|                                                                        | RadStd      |        | 0.18  | 0.35  | 0.39  | 0.34  | 0.50  | 0.57  | 0.39  | 0.38  | 0.33  | 0.31  | 0.19  | 0.12  |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | +0.50 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +122    | N/A     |
| (47) Regional (5 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +122    | N/A     |

Nomenclature: See separate page